MUHAMMAD AHMAD
FA23-BCE-113

Data Communications and Computer Networks

ASSIGNMENT # 01

## Scenario 1: Performance analysis of a wired Monitoring link:

Data:

| | |
|---|---|
| Distance | 500 km |
| Packet size | 1000 bytes |
| Data rate | 500 kbps |
| Available bandwidth | 100KHz |
| Signal to noice ratio | 20dB |
| Propogation speed | $2 \times 10^8$ m/s |
| no. of routers | 4 |
| Quieing delay/router | 5 µs |
| Processing delay/router | 2 µs |
| Transmitter power | 50 W |
| Receiver power | 40 W |

(a) transmission delay: $d_{trans}$

$$d_{trans} = \frac{\text{Packet size}}{\text{Data rate}}$$

$$= \frac{1000 \text{ bytes}}{500 \text{ kbps}} = \frac{8000 \text{ bits}}{500 \times 10^3 \text{ bps}}$$

$$d_{trans} = \frac{8000}{500{,}000}$$

$$\boxed{d_{trans} = 0.016 \text{ sec} = 16 \text{ ms}}$$

(b) Total end to end delay:

$$d_{total} = d_{trans} + d_{prop} + \text{no. of routers}\,(d_{queue} + d_{proc})$$

$$=$$

- $d_{prop} = \dfrac{\text{Distance}}{\text{Propogation delay}}$

$$= \frac{500\ km}{2\times10^{8}\ m/s}$$

$$= \frac{500{,}000\ m}{200{,}000{,}000\ m/s}$$

$$\boxed{d_{prop} = 0.0025 = 2.5\ ms}$$

Now:

$$d_{total} = 16\ ms + 2.5ms + 4\,(5\mu s + 2\mu s)$$

$$= 16\ ms + 2.5ms + 28\ \mu s$$

$$= 16\ ms + 2.5ms + 0.028\ ms$$

$$\boxed{d_{total} = 18.528\ ms = 0.018528 sec}$$

(c) Maximum achivable data rate using shannon capacity:

$$C = B\ \log_2(1+SNR)$$

$$SNR = 10^{SNR_{dB}/10}$$

$$SNR = 10^{20/10}$$

$$SNR = 100$$

$$C = 100K \cdot \log_2(1+100)$$

$$= 100{,}000 \times \log_2(101)$$

$$= 100\,000 \times 6.658$$

$$\boxed{C = 665800\ bps = 0.6658\ Mbps}$$

(d) minimum bandwidth for Binary signaling:

Given data rate $C = 500 kbps$, binary signaling $L = 2$

$$C = 2B \log_2 L$$

$$C = 2B \log_2 2$$

$$C = 2B$$

$$B = \frac{C}{2}$$

$$B = \frac{500 kbps}{2}$$

$$\boxed{B = 250 kbps = 250{,}000 bps}$$

(e) Attenuation of signal:

$$\text{Attenuation} = 10 \log_{10} \left( \frac{P_{received}}{P_{transmitted}} \right)$$

$$= 10 \cdot \log_{10} \left( \frac{40}{50} \right)$$

$$= 10 \cdot \log_{10} (0.8)$$

$$= 10 \cdot (-0.0969)$$

$$\boxed{\text{Attenuation} = -0.969 dB}$$

Scenario 2: Design of a wireless Surveillance Communication Link:

Given Data:

| | |
|---|---|
| Distance | 50km |
| Available bandwidth | 5 MHz |
| Signal to noise ratio | 30dB |
| Target data rate | 20 Mbps |
| packet size | 125000 bytes = 1 MBits |
| Propagation speed | $3 \times 10^8$ m/s |
| no of relay nodes | 2 |
| Queing delay / node | 1 ms |
| Processing delay / node | 0.5ms |
| max. toterable delay | 10 ms |

(a) Maximum achivable data rate using shannon Capacity:

$C = B \log_2 (1 + SNR)$ $\quad$ $SNR = 10^{SNR_{dB}/10}$

$= 10^{30/10} = 10^3$

$C = 5M \cdot \log_2 (1 + 1000)$ $\quad$ $SNR = 1000$

$C = 5 \times 10^6 \times \log_2 (1001)$

$C = 5 \times 10^6 \times 9.97$

$C = 49.8 \times 10^6$ bps $= 49.8$ Mbps

(b) minimum no of signal levels:

$$C = 2B \log_2 L$$

$$\frac{C}{2B} = \log_2 L$$

$$2^{(C/2B)} = L$$

$$\frac{C}{2B} = \frac{20 \times 10^6}{2 \times 5 \times 10^6}$$

$$\frac{C}{2B} = \frac{20}{10} \times \frac{10^6}{10^6} = 2$$

$$2^{C/2B} = L$$

$$L = 2^2$$

$$\boxed{L = 4}$$

(C) Minimum SNR required for target data rate

for target data rate $C = 20$ MBPS

from shannon capacity:

$$C = B \log_2 (1 + SNR)$$

$$\log_2 (1 + SNR) = \frac{C}{B}$$

$$2^{C/B} = 1 + SNR$$

so,

$$\frac{C}{B} = \frac{20 \times 10^6}{5 \times 10^6} = 4$$

$$2^4 - 1 = SNR$$

$$SNR = 15$$

$SNR_{dB} = 10\log_{10}(SNR)$

$SNR_{dB} = 10 \log_{10}(15)$

$= 10 \times 1.176$

$\boxed{SNR_{dB} = 11.76\ dB}$

(d) total end to end delay:

$d_{total} = d_{trans} + d_{prop} + \text{no. of } \cancel{\text{chan}} \text{ relay nodes } (d_{queue} + d_{proc})$

$d_{trans} = \dfrac{\text{Packet size}}{\text{Data Rate}}$

$= \dfrac{1\ MB}{20\ MBps}$

$= 0.05\ s$

$d_{trans} = 50\ ms$

$d_{prop} = \dfrac{\text{Distance}}{\text{Propagation Speed}}$

$= \dfrac{50 \times 10^3\ m}{3 \times 10^8\ m/s}$

$= 0.000167\ s$

$= 0.167\ ms$

$d_{total} = 50\ ms + 0.167\ ms + 2 \times (1 \times 10^{-3} + 0.5 \times 10^{-3})$

$= 50\ ms + 0.167\ ms + 3\ ms$

$d_{total} = 53.17\ ms$

$\boxed{d_{total} = 0.053167\ sec}$

(e) delay margin:

$d_{margin} = d_{tolerable} - d_{Actual}$

$= 10\ ms - 53.17\ ms$

$\boxed{d_{margin} = -43.17\ ms}$